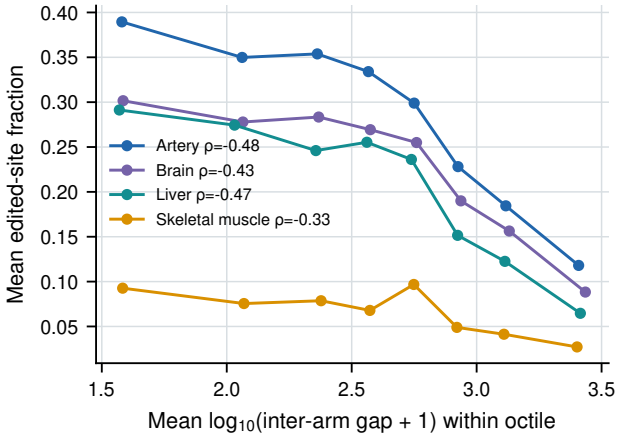
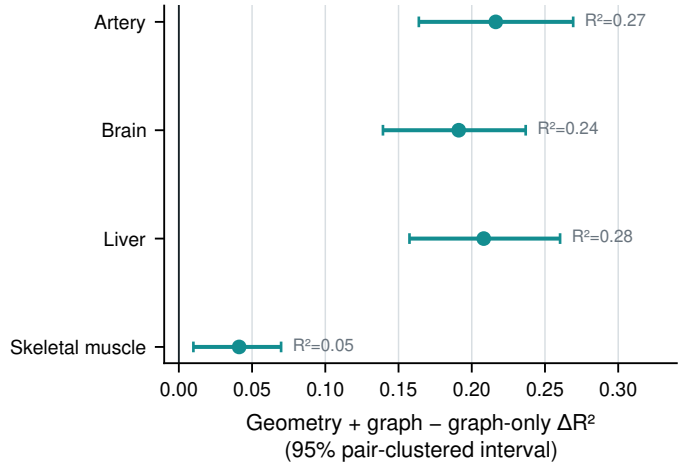
A

Evidence map: which information changes the recoverable signal?

INFORMATION LEVELS	Tested formulation	Outcome	Interpretation
LEARNING	**LEARNING & SUPERVISION**		
	Pair-first expanded supervision	● Selected	More labels; correct pair weight
	Continuous + ordinal auxiliary labels	▲ Limited	Small ranking gain
	Shared four-tissue training	▲ Limited	Transfer below preset criteria
TARGET-LOCAL	**TARGET REPRESENTATION**		
	Target/partner-aligned local view	● Selected	Passes shifted-window control
	Focal node / contact geometry*	▲ Limited	Target-position signal
	Frozen RNA-language embeddings	✖ Not supported	F1-only; no ranking gain
	Substrate-wide all-A decoder	✖ Not supported	Stopped for futility
	Supplied Bio-aware variant	▲ Limited	Features/capacity confounded
	Graph Transformer / GATv2 expansion	+ Not pursued	Not tested after local gains
SUBSTRATE / CONTEXT	**SUBSTRATE & CELLULAR CONTEXT**		
	Pair geometry / editing burden	◆ Supported†	Substrate propensity detectable
	Hierarchical engagement gate	▲ Limited	Ranking gain; unstable F1
	Bulk H3K36me3 overlap*	▲ Limited	Metric-specific screen effects
	RBP binding × bulk expression*	✖ Not supported	No consistent screen gain
	Transcript / repeat annotations*	✖ Not supported	No site-level gain
	Matched donor/cell multimodality	■ Deferred	Matched data unavailable
	Matched in vivo structure / isoforms	■ Deferred	Matched data unavailable
INTEGRATION	**INTEGRATION**		
	Fixed whole-duplex + local-augmented logits	● Selected	Complementary predictions
	Unified end-to-end architecture	✖ Not supported	Below frozen ensemble

*Pooled-score exploratory screens. Comparators and intervals: Table S2.

B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